Figure 1: Concatenating the file */dev/psaux* which represents the mouse

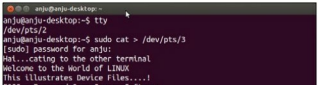

Figure 2: Concatenating to the file */dev/pts/3* which represents the second terminal

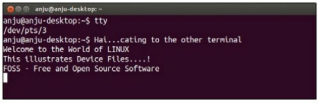

Figure 3: Displaying the concatenated lines of text on the second terminal

is an Open Document Text file. The one created using LibreOffice Impress is an Open Document Presentation file. Normally, the files of these applications have a three letter extension to the file name, and this extension reveals the type of that file. We know the extensions *odt* (Open Document Text), *odp* (Open Document Presentation), *ods* (Open Document Spreadsheet), etc. We are also quite familiar with *pdf* files (pdf stands for Portable Document File). Even a directory or folder is a special file.

Here are two methods for identifying the type of a file.

Method 1: The first character in the long listing of a file denotes its file type. To get the long listing, use the command *ls* with the *-l* option, i.e., *ls -l*

The first character in the long list and the associated file type:

- \- Ordinary file
- d Directory file
- c Character special device files (e.g., keyboard, terminal, etc)
- b Block special device files (e.g., hard disk and its partitions)
- l Symbolic link files
- p Named pipes
- s Semaphore

The following are a few commands and their outputs, which I tried in my system. Replace the file names/folder names in these examples with those in your system:

Ordinary/regular file:

```
ls   -l   ivn.odt
-rw-rw-r-- 1 ctstaff ctstaff 61110 Nov 27 14:17 ivn.odt
```

Directory file:

```
ls   -ld  CFRD
drwx------ 16 ctstaff ctstaff 4096 Nov 10 13:55 CFRD
```

Symbolic link file:

```
ls -l  lnfile
lrwxrwxrwx 1 ctstaff ctstaff 5 Nov  10 15:40 lnfile -> sqr.c
```

Block special device file:

```
ls   -l  /dev/sda
brw-rw---- 1 root disk 8, 0 Nov 30 13:21 /dev/sda
```

Character special device file:

```
ls   -l  /dev/tty1
crw-rw---- 1 root tty 4, 1 Nov 30 13:21 /dev/tty1
```

Character special device file:

```
ls  -l  /dev/psaux
crw------- 1 root root 10, 1 Nov 30 13:21 /dev/psaux
```

Character special device file:

```
ls  -l  /dev/pts/0
crw--w---- 1 root tty 136, 0 Nov 30 13:26 /dev/pts/0
```

Here, the file type Ordinary file or Regular file denotes all the data files (those created by applications such as LibreOffice Writer, Impress, gedit, etc) as well as executables (programs). Thus, by long listing these files, we cannot distinguish between them. The best way to identify the exact file type is to use the command *file*. Method 2 illustrates this command.

Method 2: The *file* utility determines the file type. Linux does not use extensions to determine the file type. Hence, this command is the best option to determine the file type. The following are a few examples I tried in my Linux system. Replace the file names in these examples with those in your system:

```
file   j.c

     j.c: C source, ASCII text

file  index.jpeg

index.jpeg: JPEG image data, JFIF standard 1.01

file  SysADM.pdf

     SysADM.pdf: PDF document, version 1.4

file  SysADM.odp

     SysADM.odp: OpenDocument Presentation
```